

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
ENTERPRISE JAVA

Practical – VI

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AIM(A): To develop a simple EJB application that counts how many times a servlet/page has been visited using an EJB Session Bean.

index.jsp

```
<!DOCTYPE html>
<html>
<head>
  <title>Hit Count Demo</title>
</head>
<body>
<form action="ShowHitCountServlet" method="post">
  <input type="submit"
    value="Take Hit Count demo with EJB Session Singleton Bean"/>
</form>
</body>
</html>
```

STHitCountBean.java

```
package EJBPract;
import javax.ejb.Singleton;
@Singleton
public class STHitCountBean {
  private int count;
  public synchronized int incrHitCount() {
    return count++;
  }
}
```

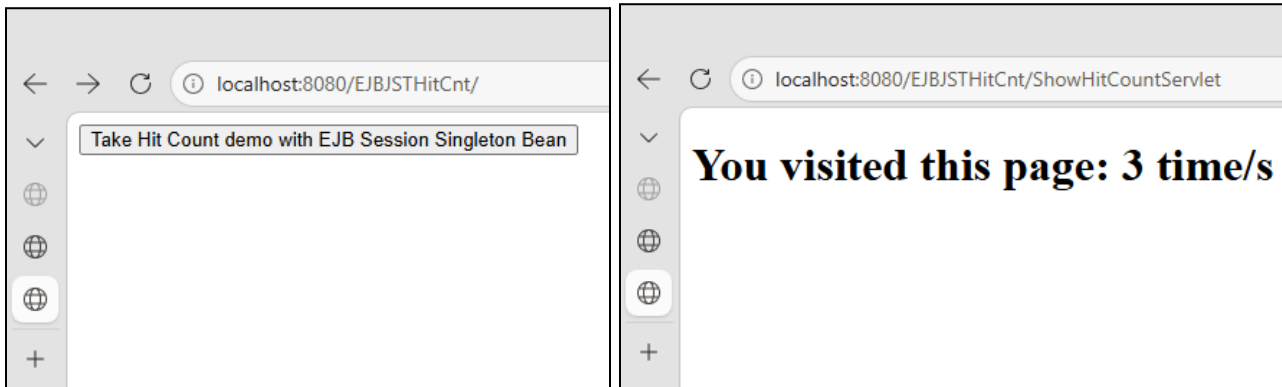
ShowHitCountServlet.java

```
package servlet;
import EJBPract.STHitCountBean;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "ShowHitCountServlet",
  urlPatterns = {"/ShowHitCountServlet"})
public class ShowHitCountServlet extends HttpServlet {
  @EJB
  STHitCountBean cntBean;
  protected void processRequest(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
```

```

response.setContentType("text/html;charset=UTF-8");
try (PrintWriter out = response.getWriter()) {
    out.println("<!DOCTYPE html>");
    out.println("<html>");
    out.println("<head>");
    out.println("<title>Servlet ShowHitCountServlet</title>");
    out.println("</head>");
    out.println("<body>");
    out.println("<h1>You visited this page: "
        + cntBean.incrHitCount()
        + " time/s</h1>");
    out.println("</body>");
    out.println("</html>");
}
}
@Override
protected void doGet(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
@Override
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
}

```

Output:

AIM(B): Develop a visitor statistics application using Message Driven Bean (Stateless Session Bean).

index.jsp

```

<%@page import="javax.jms.JMSEException"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.QueueConnectionFactory"%>
<%@page import="javax.jms.QueueConnection"%>
<%@page import="javax.jms.Session"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>

```

```

<meta http-equiv="Content-Type"
      content="text/html; charset=UTF-8">
<title>Message Driven Bean</title>
</head>
<body>
<h1>Welcome To Message Driven Bean</h1>
<%
    QueueConnectionFactory connFact = null;
    Queue queue = null;
    QueueConnection conn = null;
    Session mysession = null;
    MessageProducer msgProducer = null;
    TextMessage msg = null;
    try {
        InitialContext ctx = new InitialContext();
        queue = (Queue) ctx.lookup("jms/Queue");
        connFact = (QueueConnectionFactory)
            ctx.lookup("jms/QueueFactory");
        conn = connFact.createQueueConnection();
        mysession = conn.createSession(
            false,
            Session.AUTO_ACKNOWLEDGE
        );
        msgProducer = mysession.createProducer(queue);
        msg = mysession.createTextMessage();
        msg.setText(request.getRemoteAddr());
        msgProducer.send(msg);
        conn.close();
    } catch (JMSEException e) {
        out.println("JMS Exception: " + e.getMessage());
    } catch (Exception e) {
        out.println("Exception: " + e.getMessage());
    }
}%>
</body>
</html>

```

VStatelessBean.java

```

package EJBPract;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
    private Connection conn = null;
    private Statement stmt = null;
    @PostConstruct
    public void connect() {

        try {

```

```
Class.forName("com.mysql.jdbc.Driver");
String connString =
    "jdbc:mysql://localhost/reg";
conn = DriverManager.getConnection(
    connString,
    "root",
    "root"
);
System.out.println("Connection Success.");
} catch (ClassNotFoundException e) {
    System.err.println(
        "MySQL Driver not found: "
        + e.getMessage()
    );
} catch (SQLException e) {
    System.err.println(
        "Connection failed: "
        + e.getMessage()
    );
}
}
}
@PreDestroy
public void disconnect() {
    try {
        if (conn != null) {
            conn.close();
        }
        System.out.println("Connection closed.");
    } catch (SQLException e) {
        System.err.println(
            "Not able to close connection: "
            + e.getMessage()
        );
    }
}
}
public void addVisitor(String host) {
    try {
        stmt = conn.createStatement();
        String query =
            "INSERT INTO visitorstat "
            + "(host, visitcount) "
            + "VALUES ("
            + host
            + ", 1)";
        stmt.executeUpdate(query);
    } catch (SQLException e) {
        try {
            stmt = conn.createStatement();
            String query =
                "UPDATE visitorstat "
                + "SET visitcount = visitcount + 1 "
                + "WHERE host = "
                + host
```

```

        + "";
        stmt.executeUpdate(query);
    } catch (SQLException ex) {
        System.err.println(
            "Cannot execute update: "
            + ex.getMessage()
        );
    }
}
}
}
}

```

MsgDrvnBean.java

```

package EJBPract;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.jms.JMSException;
import javax.jms.Message;
import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(
    activationConfig = {
        @ActivationConfigProperty(
            propertyName = "acknowledgeMode",
            propertyValue = "Auto-acknowledge"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationType",
            propertyValue = "javax.jms.Queue"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationLookup",
            propertyValue = "jms/Queue"
        )
    }
)
public class MsgDrvnBean implements MessageListener {
    @EJB
    private VStatelessBean vs;
    @Override
    public void onMessage(Message message) {
        if (message instanceof TextMessage) {
            try {
                TextMessage msg = (TextMessage) message;
                String host = msg.getText();
                System.out.println(
                    "Message received from host: " + host
                );
                vs.addVisitor(host);
            } catch (JMSException e) {
                System.err.println(
                    "JMS Exception: " + e.getMessage()
                );
            }
        }
    }
}

```

```

    }
}
}
}

```

Output:

Field	Type	Null	Key	Default	Extra
visitdate	timestamp	NO		NULL	
host	varchar(45)	NO	PRI	NULL	
visitcount	int	YES		NULL	

3 rows in set (0.00 sec)

AIM(C): Develop a JSP application to display data from a MySQL Database using JDBC.

index.jsp

```

<%@page import="javax.naming.InitialContext"%>
<%@page import="EJPracts.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
  <meta http-equiv="Content-Type"
    content="text/html; charset=UTF-8">
  <title>Marks Entry</title>
</head>
<body>
<h2 style="text-align:center;">Enter Details</h2>
<form method="post">
  <table align="center">
    <tr>
      <td>Enter Student Name</td>
      <td>
        <input type="text" name="stuName">
      </td>
    </tr>
    <tr>
      <td>Enter Subject 1 Marks</td>
      <td>
        <input type="text" name="sub1">
      </td>
    </tr>
    <tr>
      <td>Enter Subject 2 Marks</td>
      <td>
        <input type="text" name="sub2">
      </td>
    </tr>
  </table>

```

```
<tr>
  <td>Enter Subject 3 Marks</td>
  <td>
    <input type="text" name="sub3">
  </td>
</tr>
<tr>
  <td></td>
  <td>
    <input type="submit"
      name="insertMarks"
      value="InsertMarks">
  </td>
</tr>
</table>
</form>
<%
if (request.getParameter("insertMarks") != null) {
  try {
    InitialContext ctx = new InitialContext();
    MarksEntryBean marksEntry =
      (MarksEntryBean) ctx.lookup(
        "java:global/EJBMarksEntry/MarksEntryBean"
      );
    String name = request.getParameter("stuName");
    String marks1 = request.getParameter("sub1");
    String marks2 = request.getParameter("sub2");
    String marks3 = request.getParameter("sub3");
    marksEntry.addMarks(
      name,
      marks1,
      marks2,
      marks3
    );
    out.println("<h3 style='text-align:center;'>"
      + "Marks Inserted Successfully"
      + "</h3>");
  } catch (Exception e) {
    out.println("<h3 style='text-align:center;'>"
      + "Error: " + e.getMessage()
      + "</h3>");
    e.printStackTrace();
  }
}
%>
</body>
</html>
```

MarksEntryBean.java

```
package EJPracts;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
```

```
import javax.ejb.Stateful;
@Stateful
public class MarksEntryBean {
    private Connection conn = null;
    private Statement stmt = null;
    public void createConnection() {
        try {
            Class.forName("com.mysql.jdbc.Driver");
            String conStr =
                "jdbc:mysql://localhost/reg";
            conn = DriverManager.getConnection(
                conStr,
                "root",
                "root"
            );
            System.out.println("Connection Success.");
        } catch (ClassNotFoundException e) {
            System.err.println(
                "MySQL Driver not found: "
                + e.getMessage()
            );
        } catch (SQLException e) {
            System.err.println(
                "Database connection failed: "
                + e.getMessage()
            );
        }
    }
    public void addMarks(
        String stuName,
        String marks1,
        String marks2,
        String marks3) {
        createConnection();
        if (conn == null) {
            System.err.println(
                "Database connection is NULL."
            );
            return;
        }
        try {
            stmt = conn.createStatement();

            String query =
                "INSERT INTO marks "
                + "(stuName, sub1, sub2, sub3) "
                + "VALUES ("
                + stuName + ","
                + marks1 + ","
                + marks2 + ","
                + marks3 + ")";
            stmt.executeUpdate(query);
            System.out.println(
```



```
        "Marks inserted successfully."
    );
} catch (SQLException e) {
    System.err.println(
        "Failed to insert marks: "
        + e.getMessage()
    );
} finally {
    try {
        if (stmt != null) {
            stmt.close();
        }
        if (conn != null) {
            conn.close();
        }
    } catch (SQLException e) {
        System.err.println(
            "Close error: "
            + e.getMessage()
        );
    }
}
}
```

Output:**Enter Details**

Enter Student Name

Enter Subject 1 Marks

Enter Subject 2 Marks

Enter Subject 3 Marks

Marks Inserted Successfully